

MEHEK MANDAL

Dubai, UAE / Los Angeles, CA | +1 213 574 6270 | mehekman@usc.edu | linkedin.com/in/mehekmandal

AI agent engineer and founder. Sole engineer of the G42 Intelligence Agent Platform submitted with this application: designed, built, benchmarked, and deployed end to end.

SELECTED SYSTEMS

G42 Intelligence Agent Platform · this submission 2026

- Governed agent chassis (LangGraph, FastAPI) serving three enterprise agents (Finance, Human Capital, Strategic Sourcing) from one live deployment: hash-addressed config snapshots, replayable checkpoint audit trail, OpenTelemetry, Presidio PII redaction, graph-level JWT RBAC, per-task budget rails.
- Oracle-anchored eval harness: 135/135 run reliability, escalation precision 1.00 on all three agents, 10/10 identity-pair bias parity (Human Capital), 10/10 adversarial attack classes defended.
- Governed self-improvement loop (eval, diagnose, propose, shadow eval, human gate); two promoted revisions in the CHANGELOG with their benchmark deltas.

Nucleus at Elemental Growth, AI agency · Founder 2025 - Present

- Built and operate Nucleus, the agency's production analytics and client-delivery pipeline supporting growth engagements with Perplexity, Gamma, and Chess.com; its 15+ integration connector pattern (fetch/normalize/cache) is the connector layer of the G42 platform.

Dial · LLM routing and evaluation 2026

- Eval harness with a 9-metric scorecard: oracle labeling via batch inference, LLM-as-judge with anti-bias controls, cache-aware cost modeling. Adapted as the G42 platform's benchmark methodology.

EXPERIENCE

Traeco, AI agent cost infrastructure · Co-Founder and CEO Feb 2026 - Jun 2026

- Ran 50+ enterprise discovery interviews on agent cost and governance failure modes; sole engineer of the Python SDK, LangChain instrumentation, and LLM-as-judge eval pipeline quantifying cost-quality deltas before deployment; live with 3 enterprise design partners in 6 weeks.
- Built config-layer observability (agent configuration captured at onboarding, not trace-only), now the audit-snapshot pattern in the G42 platform.

Tonee, AI texting tone detector · Founder Sep 2025 - Present

- Real-time tone detection for 100+ active users; fine-tuned on 8,300+ annotations from 47 user interviews, workplace tone accuracy 78% to 89%; Core ML migration cut latency 2.3s to 0.1s.

SoFi · Product Management Intern Feb 2025 - May 2025

- Analyzed an 80K monthly sign-up onboarding funnel; identified the 40% drop-off step; recommendations contributed to a 15% retention improvement.

EDUCATION

University of Southern California, Viterbi School of Engineering, Los Angeles, CA May 2027

BS Computer Science and Business Administration. GPA 3.89/4.0. USC Lava Lab startup accelerator.

SKILLS

Python, TypeScript, LangGraph, LangChain, FastAPI, Docker, OpenAI API, Anthropic SDK, agent orchestration, tool-calling patterns, LLM evaluation and LLM-as-judge methodology, fine-tuning (LoRA/QLoRA), RAG and vector search, OpenTelemetry, SQL